

Slopes and local change

Project: Image-pattern Network Investigation

1. Your mission

Your learning target

I can calculate an average rate of change over a stated interval.

I can estimate a local slope with nearby loss values.

I can explain the limits of a local direction clue.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Loss functions. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Calculate an average rate of change over a stated interval

Estimate a local slope with nearby loss values

Explain the limits of a local direction clue

Retrieve what you already know

For one example with prediction 4 and target 2, what is the squared error?

My starting explanation

This week's end goal

By the end, I can explain slopes and local change using a traced example and a stated limitation.

2. Notice the evidence

Let a one-example model predict w and let its target be 2. Its squared loss is $L(w) = (w - 2)$ squared. Here w is the parameter we vary, not a new data example. L is a nonnegative function of that parameter.

w · Prediction minus target · $L(w)$

w : 0

Prediction minus target: -2

$L(w)$: 4

w : 1

Prediction minus target: -1

$L(w)$: 1

w : 2

Prediction minus target: 0

$L(w)$: 0

w : 3

Prediction minus target: 1

$L(w)$: 1

w : 4

Prediction minus target: 2

$L(w)$: 4

From $w = 0$ to $w = 1$, the loss changes by -3 while w changes by 1. The average slope is -3. From $w = 1$ to $w = 2$ it is -1. The curve does not have one fixed slope everywhere.

List the given facts and one fact that is not supplied.

3. See how it works

Near $w = 1$, use two symmetric nearby values: 0.9 and 1.1. Their losses are 1.21 and 0.81. Change in loss divided by change in w gives $(0.81 - 1.21) / (1.1 - 0.9) = -2$. This symmetric estimate describes change near the centre $w = 1$.

```
def loss(w):  
    return (w - 2) ** 2  
  
def local_slope(w, step):  
    if step <= 0:  
        raise ValueError("Step must be positive")  
    return (loss(w + step) - loss(w - step)) / (2 * step)  
  
for w in [1, 2, 3]:  
    print(w, round(local_slope(w, 0.1), 6))  
  
# Slopes near these centres: -2, 0, 2
```

For this quadratic, the symmetric estimate equals the exact slope $2(w - 2)$ in exact arithmetic. Numerical estimates for other functions are generally approximate. Smaller steps can reduce some approximation errors, but extremely small steps can suffer from computer rounding.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

At centre $w = 0$, use step 0.5. Calculate $L(-0.5)$, $L(0.5)$ and the symmetric slope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Move from 0 all the way to 5.

Prediction: Loss becomes 9, exceeding the starting 4.

Reason: A negative local slope suggests increasing a locally, not that every positive jump decreases loss.

5. Words that unlock the idea

Word · Meaning

Word: Slope

Meaning: Change in output divided by change in input over a stated interval

Word: Local change

Meaning: Change considered near a particular input or parameter value

Word: Step size

Meaning: The amount by which an input or parameter is changed

Word: Stationary point

Meaning: A point where a differentiable function has zero local slope

Use two terms to explain a step in this week's example.

6. Follow a worked example

Compare slope clues on the same loss curve

Current w · Local slope · Nearby downhill direction

Current w: 1

Local slope: -2

Nearby downhill direction: Increase w a little

Current w: 2

Local slope: 0

Nearby downhill direction: No first order direction

Current w: 3

Local slope: 2

Nearby downhill direction: Decrease w a little

These are local clues for $L(w) = (w - 2)^2$. At $w = 2$ the displayed function has its minimum; zero slope does not establish a minimum for every possible function.

At $w = 3$, use 2.9 and 3.1. Calculate both losses and the symmetric slope. Which small change should lower loss here?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time.

Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Slopes and local change

Use a local slope to propose a small move rather than claiming every move helps.

The facts we will use

Toy loss $J(a)=(a-2)^2$. At $a = 0$, use symmetric step 0.5.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

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Slopes and local change

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Evaluate nearby points

$J(0.5)=2.25$ and $J(-0.5)=6.25$.



2. Estimate slope

$\text{Slope}=(2.25-6.25)/(0.5-(-0.5))=-4$.



3. Check small move

Move to $a = 0.25$: loss 3.0625, below starting loss 4.

CHANGE ONE THING

Move from 0 all the way to 5.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Evaluate nearby points

What happens and why: $J(0.5)=2.25$ and $J(-0.5)=6.25$.

Step: 2. Estimate slope

What happens and why: $\text{Slope}=(2.25-6.25)/(0.5-(-0.5))=-4$.

Step: 3. Check small move

What happens and why: Move to $a = 0.25$: loss 3.0625, below starting loss 4.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A slope describes how the loss changes near the current parameter. The symmetric calculation compares equal small moves on either side. Its negative result suggests a small increase may help at zero. It does not promise every increase helps: jumping to five passes the optimum at two and produces a larger loss. Check the new loss after each proposed step.

What this example does not establish

A finite difference approximates local change; numerical precision and nonsmooth points need care in more complex objectives.

7. Find and repair a mistake

A negative slope at $w = 1$ suggests that a sufficiently small increase reduces this smooth loss locally. Moving to 1.1 lowers loss from 1 to 0.81. A jump all the way to 4 raises loss to 4. The initial slope cannot justify a step of any size.

A zero slope is also limited evidence. The nonnegative function $F(w) = (w^2 - 1)^2$ has zero local slope at $w = 0$, yet $F(0) = 1$ and $F(0.1) = 0.9801$. Small moves on either side lower that function. A flat first-order clue can occur at a point that is not a minimum.

Locate the earliest unsupported step

Starting at $w = 3$, compare loss after moving to 2.9 and after moving to -1. Both moves decrease w .

State one repair and the checks you would rerun.

8. Practise together

For $L(w) = (w - 2)^2$, calculate the average slope from $w = 0$ to $w = 2$. Compare it with the local slope at $w = 2$.

At $w = 3$, use 2.9 and 3.1. Calculate both losses and the symmetric slope. Which small change should lower loss here?

Explain your prediction to a partner before comparing results.

Practise with less help: Check a symmetric estimate

Use workbook W2 for the connected example “Slopes and local change”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Slopes and local change. Proposed explanation: “Divide by the step only”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

At centre $w = 0$, use step 0.5. Calculate $L(-0.5)$, $L(0.5)$ and the symmetric slope.

```
def loss(w):  
    return (w - 2) ** 2  
  
def local_slope(w, step):  
    if step <= 0:  
        raise ValueError("Step must be positive")  
    return (loss(w + step) - loss(w - step)) / (2 * step)  
  
for w in [1, 2, 3]:  
    print(w, round(local_slope(w, 0.1), 6))
```

Slopes near these centres: -2, 0, 2

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

For $L(w) = (w - 2)^2$, calculate the average slope from $w = 0$ to $w = 2$. Compare it with the local slope at $w = 2$.

Starting at $w = 3$, compare loss after moving to 2.9 and after moving to -1. Both moves decrease w .

For $F(w) = (w^2 - 1)^2$, compare $F(0)$ and $F(0.1)$. Does zero local slope at zero prove a minimum?

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled slopes and local change evidence sheet to Image-pattern Network Investigation. Show how you can calculate an average rate of change over a stated interval. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

Always record the function, centre, nearby values and denominator used in a slope estimate. An average slope across a wide interval is not automatically the local slope at either endpoint. Do not compare slope magnitudes without considering parameter and loss units.

A learner sees a negative estimated slope and says any larger parameter value will improve the loss. Give a counterexample using this lesson.

Next week changes one parameter at a time in a two-parameter model and combines those local slopes into a gradient.

For $F(w) = (w^2 - 1)^2$, compare $F(0)$ and $F(0.1)$. Does zero local slope at zero prove a minimum?

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Compare slope clues on the same loss curve

Current w · Local slope · Nearby downhill direction

Current w : 1

Local slope: -2

Nearby downhill direction: Increase w a little

Current w: 2

Local slope: 0

Nearby downhill direction: No first order direction

Current w: 3

Local slope: 2

Nearby downhill direction: Decrease w a little

These are local clues for $L(w) = (w - 2)^2$. At $w = 2$ the displayed function has its minimum; zero slope does not establish a minimum for every possible function.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Estimate a new curve independently

Let $J(a) = (a + 1)^2$. At $a = 1$ use step 0.25 to find the symmetric slope. Compare $J(1)$ with $J(0.9)$, then explain why moving to -4 does not follow safely from the slope clue.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Let $J(a) = (a + 1)^2$. At $a = 1$ use step 0.25 to find the symmetric slope. Compare $J(1)$ with $J(0.9)$, then explain why moving to -4 does not follow safely from the slope clue.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.